

The Best Smartphones and Tablets

A BUYERS' GUIDE

Ultra-portable devices have taken the market by storm, eating into the legacy PC market. With the increasing popularity of Android as a platform, vendors are churning out a plethora of devices to choose from. This article aims to guide you on the best available.



Android has taken everyone by storm—I still remember trying out the first beta release on a virtual machine. No one believed it to be strong, but perhaps the backing of an Internet giant has its own benefits. Being open source (kind of), the Android project gained traction and within years became a real competitor to the then very strong iOS. In this article, we will be looking at devices that feature some flavour (version) of Android. But before we get to the devices, let's take a quick look at the innards.

Android is an ARM-centric OS. Even though it has now added support for MIPS and x86 instruction sets, its 'first love' is still very much the ARM instruction set and all Android releases (AOSP builds) are geared towards ARM processors.

The ARM processors jargon buster

ARM (Advance RISC Machines) is a UK company that built RISC processors, which it sells under its name. Almost all smartphones and low-power computing devices use ARM or some sort of RISC processor (e.g., MIPS) inside. ARM doesn't

actually sell microprocessors—it licenses its technology to other companies as a result companies can either drop the vanilla (non-modified) ARM Core in their SoC or create their own processors using ARM designs as the blueprint. Of course, all companies have to pay royalty to ARM.

SoC: The System on a Chip is a package that has the processing core (CPU), graphics core (GPU), memory controller, baseband and host of other controllers that are needed to process data, fabricated onto a single piece of silicon. SoCs are efficient and reduce memory overhead. All mobile devices come with an SoC from one manufacturer or the other, which may have either an ARM/MIPS or x86 core. Companies like Qualcomm, Texas Instruments, ST-Ericsson, Apple and nVidia built their SoCs using ARM processor designs.

The ARM nomenclature can confuse you easily. Unlike the x86 core/instructions, which are easy to understand and differentiate, ARM makes the job a bit tougher. To make things easier, I'll segregate the ARM core into three segments, as shown below.